Torque-speed and efficiency report

Example SPM 8p/12s. Built on demand from the design's flux-linkage map and the drive limits set on the Torque-speed tab, with the same calculation the tab uses. The run's own report records the single solved operating point.

Key results

Peak torque	22.294 N m
Base speed	1436.805 rpm
Peak power	3.464 kW
Top speed reached	2067.608 rpm
Best efficiency	96.542%

Status

Qualified outputs pass a benchmark gate against a reference; experimental outputs have none yet.

Torque-speed envelope	Experimental: Not yet compared with measured torque-speed data. Interpolated from the flux-linkage map with sinusoidal current; inverter drops and dead time are ignored.
Efficiency map	Experimental: Not yet compared with measured efficiency data. MTPA control: the smallest current for each torque within the voltage limit, interpolated from the flux-linkage map; torque is electromagnetic. Copper loss at the winding temperature; iron loss scaled from the map's reference speed, hysteresis with frequency and eddy current with its square.
Map flux linkage, MTPA trajectory and inductances	Experimental: The d/q gates cover 12 operating points, not a map's full range of currents and angles, and do not bound the inductances.
Phase resistance and copper loss	Experimental: No wire is set, so the wire is sized to fill the slot to its limit: an estimate, not the resistance of a real winding. Set the wire for a resistance the phase-resistance gate covers.

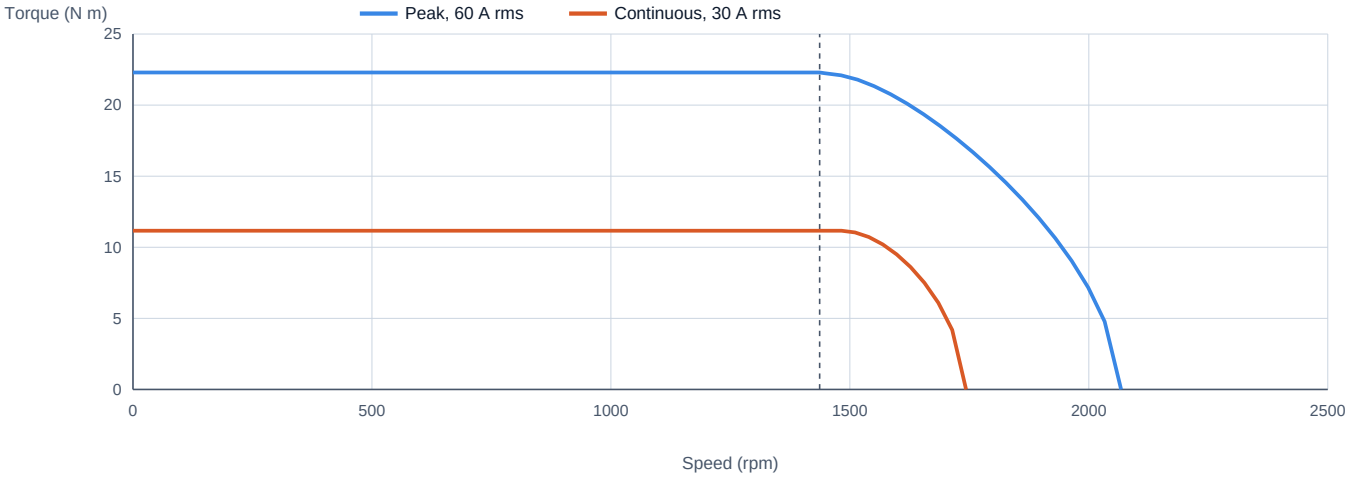
Drive inputs

Peak current	60 A rms (84.853 A peak)
Continuous current	30 A rms (42.426 A peak)
DC bus	48 V, as entered
Phase voltage limit	27.713 V peak (modulation index 1.155)
Top speed	7200 rpm
Drive control	MTPA, then field weakening (least current per torque)
Phase resistance	0.006 ohm at 20 degC, experimental
Mechanical loss	0 W
Flux-linkage map	7 currents up to 84.853 A peak x 10 angles (0 to 90 deg), 24 rotor positions each, normal mesh, standard grid; iron loss at 2400 rpm
Map scope key	333b0547c040d1c3a19a3a92ba17031c058e6395f5552cda160abdbfc92db029
coilEM	42cb14b438d206c63b170a3f204dfb3f406e2515

Torque and power against speed

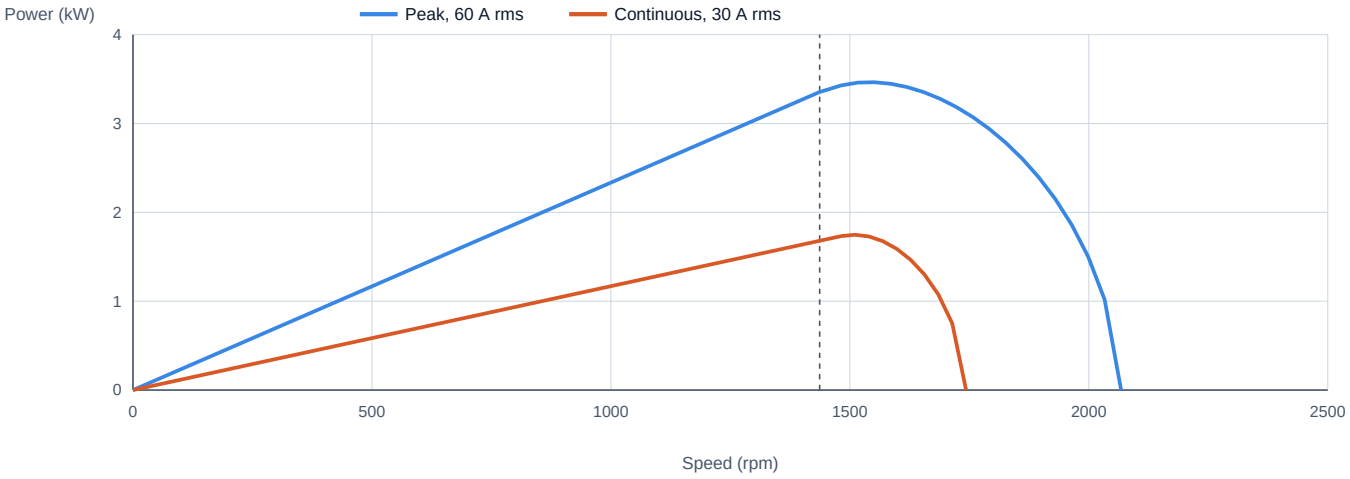
Torque against speed

Experimental



Power against speed

Experimental



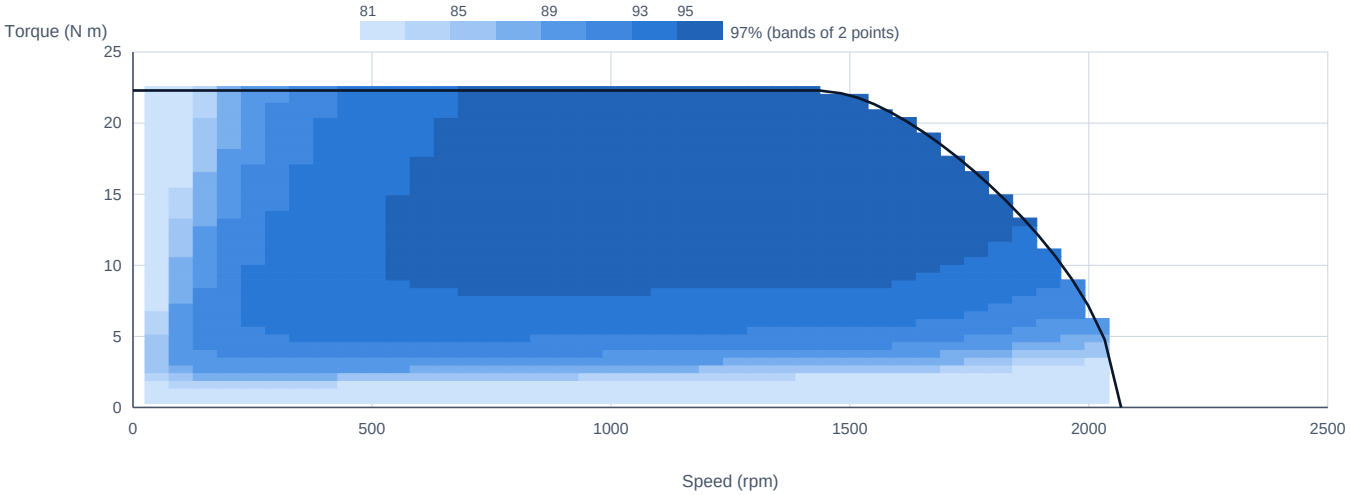
Peak curve at selected speeds (every sample is in the performance CSV)

Speed (rpm)	Torque (N m)	Power (kW)	Current (A rms)	Angle (deg)	Voltage (V peak)	Region
0	22.294	0	60	0	0.49	MTPA
206.761	22.294	0.483	60	0	4.395	MTPA
379.062	22.294	0.885	60	0	7.661	MTPA
585.822	22.294	1.368	60	0	11.581	MTPA
758.123	22.294	1.77	60	0	14.847	MTPA
964.884	22.294	2.253	60	0	18.766	MTPA
1137.185	22.294	2.655	60	0	22.033	MTPA
1343.945	22.294	3.138	60	0	25.952	MTPA
1481.786	22.095	3.429	60	6.98	27.713	Field weakening
1688.547	18.541	3.279	60	33.556	27.713	Field weakening
1860.848	13.357	2.603	60	53.089	27.713	Field weakening
2067.608	0	0	60	89.993	27.713	Field weakening

Efficiency map

Efficiency map, MTPA, then field weakening

Experimental

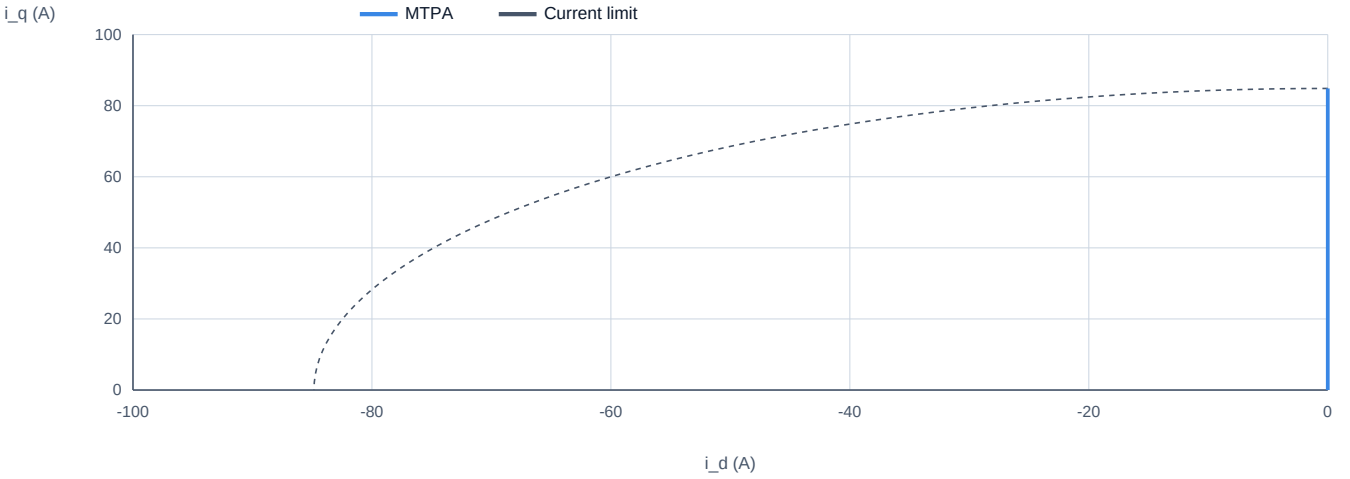


Efficiency = shaft power / input power, with shaft power = T x omega - iron loss - mechanical loss and input power = T x omega + copper loss. Cells above the peak curve are outside the drive's reach. The current and angle of every cell are in the performance CSV.

MTPA trajectory and flux linkage

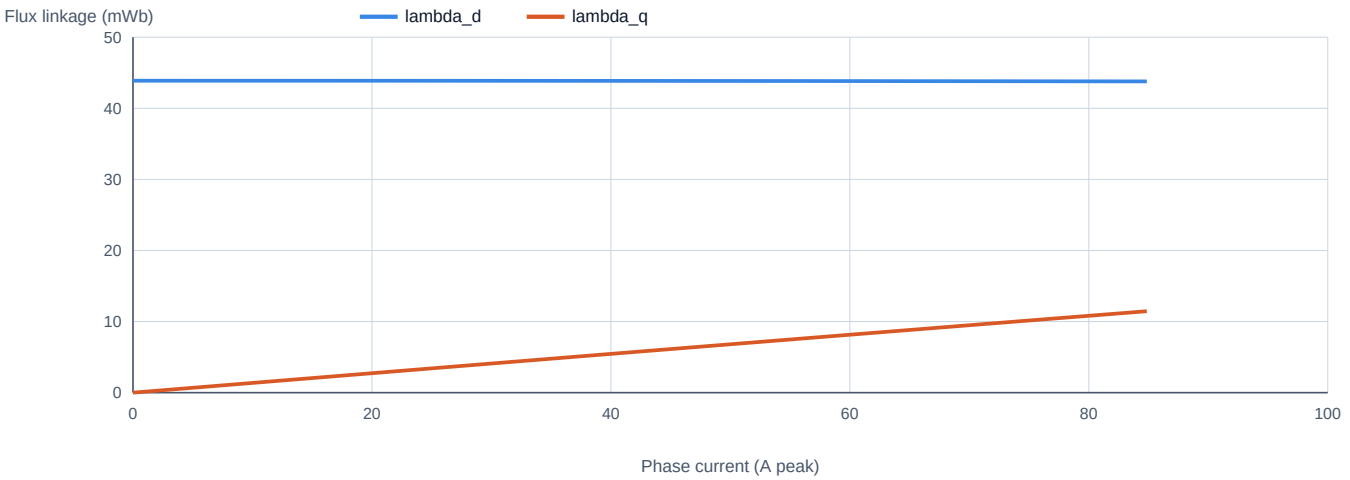
MTPA trajectory in the id-iq plane (peak A)

Experimental



Flux linkage along the MTPA trajectory

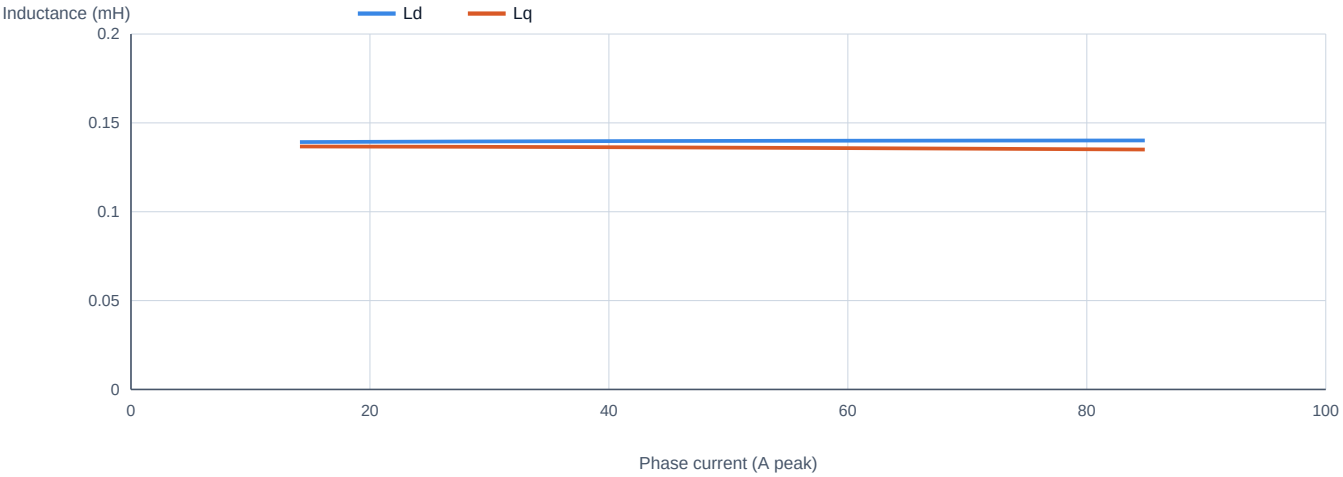
Experimental



Apparent inductance

Apparent inductance against current

Experimental



Current (A peak)	Current (A rms)	Ld (mH)	Lq (mH)
14.142	10	0.139	0.137
28.284	20	0.14	0.137
42.426	30	0.14	0.136
56.569	40	0.14	0.136
70.711	50	0.14	0.135
84.853	60	0.14	0.135

Notes

- FEM torque is 99.9-100.0% of the d/q torque across 53 map points.

- Ld from the 90 deg points, (lambda_d - lambda_pm) / i_d; Lq from the 0 deg points, lambda_q / i_q. Both are secant values and include saturation.

- The MTPA angle is read off the interpolated map, whose angles are 10 deg apart at most. At low current, where torque hardly changes with angle, it is approximate; a finer map grid gives a smoother trajectory.

- Not yet compared with measured torque-speed data. Interpolated from the flux-linkage map with sinusoidal current; inverter drops and dead time are ignored.

- Not yet compared with measured efficiency data. MTPA control: the smallest current for each torque within the voltage limit, interpolated from the flux-linkage map; torque is electromagnetic. Copper loss at the winding temperature; iron loss scaled from the map's reference speed, hysteresis with frequency and eddy current with its square.

Known limitations

- Interpolated from a flux-linkage map (Catmull-Rom over current and current angle); nothing is extrapolated beyond its currents and angles. Not yet compared with measured data.

- Steady state with sinusoidal current. Inverter voltage drops, dead time and the voltage behind the iron-loss resistance are left out.

- Torque is electromagnetic. Shaft output subtracts iron loss and the mechanical loss entered for the design.

- Iron loss is scaled from the map's reference speed: hysteresis with frequency, eddy current with its square.